Assignment 13 - Assignment on topic systemd

The objective of this assignment is to familiarize students with controlling services and daemons using systemd on Red Hat Enterprise Linux (RHEL) 9. Students will learn how to manage system services, enable/disable services, view service status, and troubleshoot common service-related issues.

You must have an access to Linux system.

Questions for this assignment

1. Write in brief about Systemd?
2. How would you manage services?
   Demonstrate how to start, stop, and restart a service using systemd commands.
   Explain how to check the status of a service to verify whether it is running or not.
3. How to Enabling and Disabling Services?
   How to enable a service to start automatically at boot time?
   Explain how to disable a service to prevent it from starting automatically at boot time?
4. Introduce the concept of service dependencies and how systemd handles them?
   Demonstrate how to view the dependencies of a service using systemctl?
5. Explain how to use systemctl and journalctl to diagnose and resolve service problems.

Solution by the Instructor:

1. Write in brief about Systemd?
   - Systemd is a system and service manager for Linux operating systems.
   - It is responsible for managing the startup process, services, and daemons.
   - Systemd uses unit files to define services, targets, and other system components.

2. How would you manage services?

   Demonstrate how to start, stop, and restart a service using systemd commands.

   Explain how to check the status of a service to verify whether it is running or not.

   - To start a service: sudo systemctl start <service_name>

- To stop a service: sudo systemctl stop <service_name>
- To restart a service: sudo systemctl restart <service_name>
- To check the status of a service: sudo systemctl status <service_name>

3. How to Enabling and Disabling Services?

   How to enable a service to start automatically at boot time?

   Explain how to disable a service to prevent it from starting automatically at boot time?

   - To enable a service: sudo systemctl enable <service_name>
   - To disable a service: sudo systemctl disable <service_name>

4. Introduce the concept of service dependencies and how systemd handles them?

   Demonstrate how to view the dependencies of a service using systemctl?

   To view service dependencies: sudo systemctl list-dependencies <service_name>

5. Explain how to use systemctl and journalctl to diagnose and resolve service problems.

   - Check the service status using systemctl status for any error messages.
   - Review the service logs using journalctl -u <service_name> to identify any issues.